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Laboratory Activity 6: Basic Actuator Control

I. Objectives

The objective of this laboratory is to control a DC motor using an ESP32, L298N motor driver, and push button. This activity demonstrates how a microcontroller controls a motor through a motor driver and how the motor responds when the button is pressed and released.

II. Materials and Components

- ESP32 Development Board
- L298N Motor Driver Module
- RF-300CA-12400 DC Motor
- Push Button
- Breadboard
- Jumper Wires
- External Motor Power Supply
- USB Cable and Laptop

III. Circuit Connections

Component	Connection	Function
Push Button	GPIO23 and GND	Controls motor operation
L298N IN1	GPIO21	Controls motor input 1
L298N IN2	GPIO22	Controls motor input 2
L298N ENA	GPIO27	Enables the motor driver channel
DC Motor	L298N OUT1 and OUT2	Receives power from the motor driver
Motor Power Supply	L298N VS and GND	Supplies power to the motor
Common Ground	ESP32 GND and L298N GND	Provides a common voltage reference

The ESP32 controls the motor driver through its GPIO pins. The motor is connected to the output terminals of the L298N instead of directly to the ESP32.

IV. Program Explanation

The program uses the ESP32 to read the push button and control the DC motor through the L298N motor driver.

1. setup()

Initializes the motor control pins as outputs and the push button as an input using INPUT_PULLUP. The motor outputs are initially set to LOW to prevent the motor from running during startup.

2. loop()

Continuously reads the state of the push button. When the button is pressed, GPIO23 reads LOW and the program commands the motor to run. When the button is released, the motor drive is disabled.

3. MOTOR_IN1 and MOTOR_IN2

These pins control the direction and operation of the DC motor. Setting IN1 to HIGH and IN2 to LOW commands the motor to rotate in one direction.

4. MOTOR_ENA

The ENA pin enables or disables the motor driver channel. Unlike the original example, which uses an nSLEEP pin, the L298N uses ENA to control its output channel.

5. Motor Driver Pin Functions

- **AIN1 / IN1:** Controls the first motor input.
- **AIN2 / IN2:** Controls the second motor input.
- **nSLEEP:** Used in the original example to enable or disable the driver. The L298N does not have this pin.
- **VM / VS:** Receives the external motor power supply.
- **Common Ground:** Connects the ESP32 and motor driver to the same ground reference.

V. Results and Observations

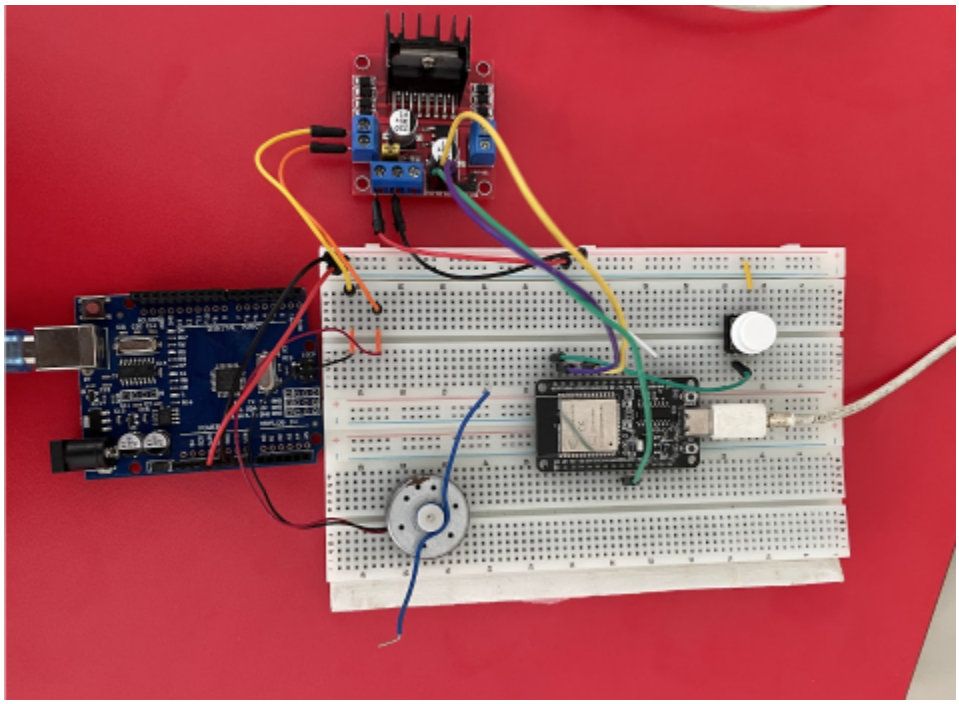
Table 1. Button-Controlled Motor Testing

Button Condition	Expected Result	Observed Result
Button released	Motor stopped	Motor stopped

Button pressed	Motor rotates	Motor rotated
Button held	Motor continues rotating	Motor continued rotating
Button released after pressing	Motor coasts to stop	Motor coasted to a stop

The DC motor was successfully operated using the motor driver. The test demonstrated that the motor could rotate when controlled by the ESP32.

Figure 1. Actual Circuit Connections



VI. Discussion

This laboratory demonstrated how an ESP32 can control a DC motor using an L298N motor driver. The motor driver allows the ESP32 to control the motor without connecting it directly to the microcontroller's GPIO pins.

The push button serves as the input, while the motor serves as the output. The ESP32 reads the button state and sends control signals to the motor driver.

The L298N was used instead of the motor driver in the original example. Therefore, the control connections were adjusted to match the available hardware.

VII. Conclusion

The laboratory demonstrated basic actuator control using an ESP32 and L298N motor driver. The DC motor was successfully operated through the motor driver. This activity helped explain how digital inputs, motor driver circuits, and external power supplies work together to control a DC motor.

VIII. Documentation

Video Documentation:
[https://drive.google.com/file/d/1S6PspRNAYud4NpF6JbZwoRldjSRxq8pZ/view?usp=drive_link]